

What are your expectations from the program you are applying to at CMU-Africa, and how will it help you achieve your professional aspirations?

AI is advancing at a fast pace, redefining and changing the construct of all we knew was possible. And yet, despite its growing population (Africa is the second largest continent), Africa is still far from taking a leading role

in the ongoing transformation. This has fueled my desire and determination to gain advanced training in the field of AI to be able to contribute my quota to building solutions that address the unique problems that the continent faces.

From the program at CMU Africa, I expect to gain an in-depth understanding of modern AI systems: how they are built, the algorithms powering them, and how they can be deployed responsibly. I am very interested in exploring the ethical side of AI, the business impacts of AI solutions, and the opportunity to work on industry-scale projects. I believe that the program blend of technical depth, professional preparation, and real-world solving is exactly what I need to grow.

My long-term aspiration is to contribute to the development of AI systems that improve Africa's technical capacity. I believe that the training and exposure I will gain at CMU Africa will serve as a launchpad for the vision of ensuring that the continent is not left out in this moment of global change.

Education, work, and life have individually prepared me for life in any place I might find myself. Growing up, I had to pick up problem-solving by readily coming up with solutions to different problems. It also introduced me to the diversity of people, which made it easy for me to work with people regardless of where they are from. This has honed me and prepared me for graduate studies at CMU Africa and any university, as it made me put on my problem-solving hat, which is essential in a graduate environment, while also maintaining the spirit of teamwork by working with different people to attain a common goal or common problem.

School in its own way has also prepared me to be able to settle in CMU Africa by ingraining in me, an approach to deal with technical problems. Seeing as my program MS EAI is one that involves starting from a problem statement and then drafting the solution through code alongside iterative testing. School prepared me for this by posing me logical questions from different perspectives, allowing me to tap into my creative side and utilising my problem-solving skills.

My school also provided a platform for me to explore the foundation of deep technical topics while having the opportunity to fully understand them. School further enhanced my teamwork skills by pairing me with teams to work on different projects with the incentive of a grade.

My experience in the corporate and professional space as both a freelancer and an employed staff has equipped me with the different skills that has ended up preparing me for CMU. While I have an independent side, I found how it is to work under the supervision of senior colleagues, who doesn't control the project but ensure that I am in accordance to what I am supposed to be doing.

Another preparation was on the side of accountability, ensuring that all I worked on, gets synced the way it should be. While also being responsible for every actions that was carried out. Actions that many have been detrimental to the intended goal or may have been highly beneficial to the achievement of the goal.

Curiosity and working to hit a goal is also something that has rubbed off, making it possible for me to work on different project independently and learn different things to boost my careers. These and various others are things that have been preparing me for life in graduate studies in CMU Africa.

Education, work, and life have individually prepared me for any situation I might find myself in. Growing up, I had to pick up problem-solving by readily coming up with solutions to various challenges. This introduced me to the diversity of people, making it easy to work with anyone regardless of their background. This has honed me for graduate studies at CMU-Africa, forcing me to put on my "problem-solving hat", essential for a graduate environment, while maintaining the

spirit of teamwork to attain a common goal.

School also prepared me to settle into CMU-Africa by ingraining in me an approach to dealing with technical problems. Since the MS EAI program involves starting from a problem statement and drafting a solution through code and iterative testing, my background is a strong fit. School prepared me for this by posing logical questions from different perspectives, allowing me to tap into my creative side. It also enhanced my teamwork abilities by pairing me on projects with the incentive of a grade.

My experience in the corporate space as both a freelancer and employee has equipped me for CMU Africa. While I have an independent side, I learned how to work under the supervision of senior colleagues who ensure I am in accordance with requirements without controlling every step. I also learned accountability, ensuring my work syncs correctly and taking responsibility for every action, whether the outcome was detrimental or beneficial to the goal.

Finally, curiosity and the drive to hit goals have rubbed off on me, allowing me to work on independent projects and learn new things to boost my career. These experiences have prepared me for life in graduate studies at CMU-Africa.

My undergraduate research in deep learning for credit card fraud detection renewed my interest in AI and showed me how powerful these systems can be when they are applied to real-world problems, which Africa has a lot of. It also made it clear that I needed to know more to be able to contribute to AI at scale.

Describe an example of a leadership, mentor or volunteer role that you held. How did it impact you and others?

A leadership position I recently held was being the president of the NYSC ICT Community Development Service, a core arm of the National Youth Service Corps responsible for the participation of corps members in their communities. This CDS is responsible for IT sensitization to schools and others.

As the president of the CDS group with 200+ members, a lot of planning and structure had to be put in place, especially for the sensitization outreach to schools and orphanage homes

Some of the impacts I got was

People management: I had to become adept at handling the reactions of people to different scenarios and handling those situations before they resulted in resentment

A huge impact I got was realising the place of sacrifice in leadership. It made me realise how much leaders sacrifice

The realization that I don't have to do everything. It was a factor that made me big on assigning, which gives me the freedom to focus on the big deal.

I was actually happy to lead the group and help others. Help

For others, The impact was empathy and involvement with what was happening

A leadership position I recently held was being the President of the NYSC (National Youth Service Corps) ICT Community Development Service (CDS) group, a core arm of the National Youth Service Corps responsible for engaging corps members in community development. Leading a group of over 200 members required significant planning and structure, particularly for our IT sensitization outreaches to local schools and orphanage homes.

This role had a profound impact on my personal growth. First, it taught me people management; I had to become skilled at navigating diverse personalities and managing reactions to different scenarios before they escalated into resentment. Second, I learned the value of sacrifice in leadership. I realized that effective leaders often prioritize the group's needs above their own comfort to ensure success. Finally, I learned the power of delegation. Realizing I did not have to do everything myself allowed me to assign tasks to capable team members. This gave me the freedom to focus on the "big picture" strategy rather than getting lost in the details.

For others, the impact was two-fold. For the communities we served, our outreaches bridged a digital gap. Students in under-resourced schools and orphanages were introduced to foundational IT

concepts, sparking a curiosity for technology that they previously did not have access to. For the corps members I led, the impact was a deepened sense of empathy and active involvement. By creating a structured environment, I transformed the group from a passive gathering into an active workforce. Members who were previously disengaged found a sense of purpose in teaching and mentoring others, allowing them to experience the fulfillment that comes from using their skills to uplift their community.

I have combined your leadership and work ethics to answer this.

To me, the values of the College of Engineering are not just ideals, but practical necessities for successful collaboration. I plan to contribute to the CMU-Africa community specifically through integrity, respect, and trust.

My professional background as both a freelancer and an employee has grounded me in integrity. I have learned that being trustworthy means more than just delivering work; it means rigorous accountability. In my previous roles, I made it a point to own every outcome—whether a project succeeded or required correction. At CMU-Africa, I will contribute to a culture of trust by being a teammate who delivers on promises, admits to technical errors early to facilitate faster learning, and gives credit where it is due.

Furthermore, my leadership experience with the NYSC ICT group taught me that respect is the foundation of innovation. Leading over 200 diverse members required me to be genuine and empathetic, ensuring that different perspectives were heard to prevent resentment. I discovered that the best ideas often come from the quietest voices when they feel safe. Within the CMU community, I will champion this inclusive mindset. I aim to be a peer who creates a safe space for dialogue in group projects and labs, trusting that our collective diversity will lead to excellent, innovative technical solutions.

The most challenging project I worked on was building the core backend logic for a real-time gaming platform featuring high-frequency games like "Crash" and "Dice." The complexity lay in handling concurrency and race conditions, ensuring that when hundreds of users attempted to "cash out" milliseconds before a game ended, the system processed every transaction accurately without financial loss or double-spending.

I used TypeScript (NestJS) for the application logic and PostgreSQL for the database. My approach started with ensuring trust; I implemented a "provably fair" algorithm using cryptographic hashes to pre-generate game outcomes, guaranteeing that results could not be manipulated.

For the implementation, I architected a real-time state machine. I utilized WebSockets to broadcast the live game multiplier to clients with low latency. To solve the concurrency challenge, I leveraged PostgreSQL transactions with strict locking mechanisms for all wallet operations. This ensured that balance deductions and winnings were atomic; either the entire transaction succeeded, or it failed safely. Additionally, I integrated Redis to cache the rapidly changing game state, which reduced database load by 30% and ensured the APIs remained responsive even during peak betting rounds.